JOINT UNIVERSITIES PRELIMINARY EXAMINATIONS BOARD

AUGUST 2022 EXAMINATIONS

OPTION – D330

BIOLOGY **Time Allowed: 3 hours**

SECTION A: MULTIPLE CHOICE QUESTIONS

Answer ALL questions in this section.

Use the OMR answer sheet provided to answer the questions.

Write and shade on your OMR Sheet the OPTION written on your question paper.

Follow the instructions on the OMR sheet.

SECTION B: ESSAY QUESTIONS

Answer FOUR Questions in ALL; ONE Question from each Course.

Ensure you read and follow all the Instructions on the cover page of the Answer Booklet.

4. The parts labelled **ii** and **iii** are together known as the

   A. piliferous layer.

   B. endodermis.

   C. vascular bundle.

   D. pericycle region.

5. Which part of a plant is usually referred to as "source"?

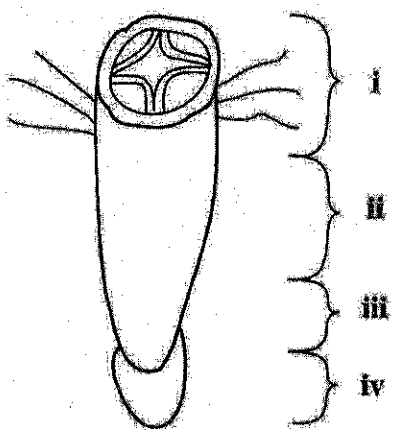
   A. Leaf.

   B. Root.

   C. Stem.

   D. Branch.

Use the diagram below to answer questions 6 and 7.



6. The part labelled **ii** is called

   A. region of maturation.

   B. region of elongation.

   C. region of cell division.

   D. root cap.

7. Which of the parts contain dense mass of protoplasm?

   A. **i**

   B. **ii**

   C. **iii**

   D. **iv**

13. The pigments phycocyanin and phycoerythrin are usually found among the algal group called
    A. Chlorophyta.
    B. Chrysophyta.
    C. Phaeophyta.
    D. Rhodophyta.

14. Which of these is inversely proportional to action of auxin?
    A. Cytokinins.
    B. Gibberellins.
    C. Abscisic acid.
    D. Ethylene.

15. Vegetative propagation using natural methods includes the following **EXCEPT**
    A. Corm.
    B. Bulb.
    C. Rhizome.
    D. Budding.

16. A vegetable seller will use one of these hormones to maintain the green colour of vegetables.
    A. Abscisic acid.
    B. Cytokinins.
    C. Auxin.
    D. Gibberellins.

17. The aspect of Zoology that deals with the behaviour of animals is known as
    A. Behavioural science.
    B. Ethology.
    C. Malacology.
    D. Herpetology.

18. The middle layer of the body wall in a diploblastic animal is termed
    A. mesoderm.
    B. ectoderm.
    C. mesoglea.
    D. endoderm.

25. The organelle responsible for the breakdown of old organelles and the digestion of bacteria is

A. ribosome.

B. nucleus.

C. mitochondria.

D. lysosome.

26. If 3 Na^+ leave a plant cell into extracellular fluid each time 2 K^+ are actively transported into the cell, how many Na^+ will be actively transported out of the cell if 5 K^+ enter into the cell?

A. 3.5

B. 3.0

C. 7.5

D. 7.0

27. Which of these statements is true about a plant cell with high water potential? It

A. gains water from its environment.

B. has high solute potential.

C. has high pressure potential.

D. losses solute to its environment.

28. Ovary, style and the stigma of a flower are found in the

A. androecium.

B. gynoecium.

C. calyx.

D. corolla.

29. The type of flowers that attach singly to either leaf axis or tip of branches are regarded as

A. inflorescence.

B. solitary flower.

C. imperfect flower.

D. complete flower.

35. The gaps in the sheath between successive Schwann cells are referred to as

A. Myelin sheath.

B. Axons.

C. Node of Ranvier.

D. Dendrites.

36. A fertilized matured ovary in plant is referred to as

A. leaf.

B. fruit.

C. root.

D. stem.

37. Which polysaccharides are branched and which are unbranched? Select the correct option.

OPTION	BRANCHED	UNBRANCHED
A.	Amylose	Cellulose
B.	Amylopectin	Cellulose
C.	Cellulose	Amylose
D.	Cellulose	amylopectin

38. Immediately after which stage in mitosis in an animal cell does the cytoplasm start to divide?

A. Anaphase.

B. Metaphase.

C. Prophase.

D. Telophase.

43. People with sickle cell anaemia have haemoglobin
    A. S and are homozygous recessive.
    B. A and are heterozygous recessive.
    C. S and are heterozygous recessive.
    D. A and are homozygous recessive.

44. Down's syndrome is as a result of chromosomal abnormalities which results in an individual with additional chromosome number of
    A. 12
    B. 21
    C. 13
    D. 8

45. Protists and bacteria are grouped into different domains because
    A. protists have a membrane-bounded nucleus, which bacterial cells lack.
    B. bacteria are not made of cells.
    C. protists eat bacteria.
    D. bacteria decompose protists.

46. The term viviparity means
    A. ability of animals to give birth to young ones alive.
    B. ability to lay eggs.
    C. ability of animals to derive food from plant.
    D. ability for parental care.

47. The organism that has a hydrostatic skeleton is
    A. *Tilapia.*
    B. *Hydra.*
    C. *Planaria.*
    D. *Lumbricus.*

SECTION B: ESSAY QUESTIONS

Answer FOUR Questions in ALL; ONE Question from each Course.

BIO 001: GENERAL BIOLOGY

1. (a) Define the following:
   i. Genes
   ii. Alleles
   iii. Homozygous organism
   iv. Heterozygous organism
   v. Dominant allele
   vi. Recessive allele
   vii. Test cross
   viii. Sex-linked genes
   ix. Rhesus
   x. Albinism. **[5 Marks]**

   (b) The following set of data represents the lengths in millimeters of 20 maize fruits harvested and measured by students during a biology practical session.
   Study the data carefully and use it to answer the following questions
   Length (mm) of Maize fruits

100	120	125	140	140	110	100	150	140	130
135	125	140	120	130	120	130	155	140	150

   Arrange the set of data from lowest to highest values. Calculate the
   i. Mean ii. Median iii. Mode iv. Range **[4 Marks]**

   (c) Define genetic variation. **[1 Mark]**

2. (a) A recent ecological study in a community, put the mass of organisms found as follow: Rats 2000g/m^2, Man 600 g/m^2, Yam 4000 g/m^2, Locust 3500 g/m^2. Represent this information using a pyramid of Biomass. **[4 Marks]**

   (b) What is trophic level of each organism? **[2 Marks]**

(c) State the function of each of the following plant tissues.

- Xylem
- Phloem
- Sclerenchyma
- Chlorenchyma **[2 Marks]**

6. (a) Define fertilization. **[1 Mark]**

(b) Differentiate between seeds and fruits of plants in a tabular form. **[4 Marks]**

(c) Describe the process of fertilization in plants. **[5 Marks]**

BIO 004: INTRODUCTORY ZOOLOGY

7. (a) Give **SIX** characteristics of the sub-kingdom protozoa. **[3 Marks]**

(b) Complete the table below by showing **ONE** example of each class of protozoa.

Groups	**Free living examples**	**Parasitic forms examples**
Sarcodina		
Mastigophora (Flagellates)		
Ciliophora (Ciliates)		
Apicomplexa (sporozoans)		

[4 Marks]

(c) Differentiate between *Paramecium* and *Amoeba*. **[3 Marks]**

8. (a) List **FOUR** reasons why transport is necessary in animals. **[2 Marks]**

(b) Itemize **SIX** functions of blood. **[3 Marks]**

(c) Differentiate between arteries and veins. **[3 Marks]**

(d) List **FOUR** functions of the skeleton. **[2 Marks]**